

Hw3: Integrity Report

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1. System Assumptions

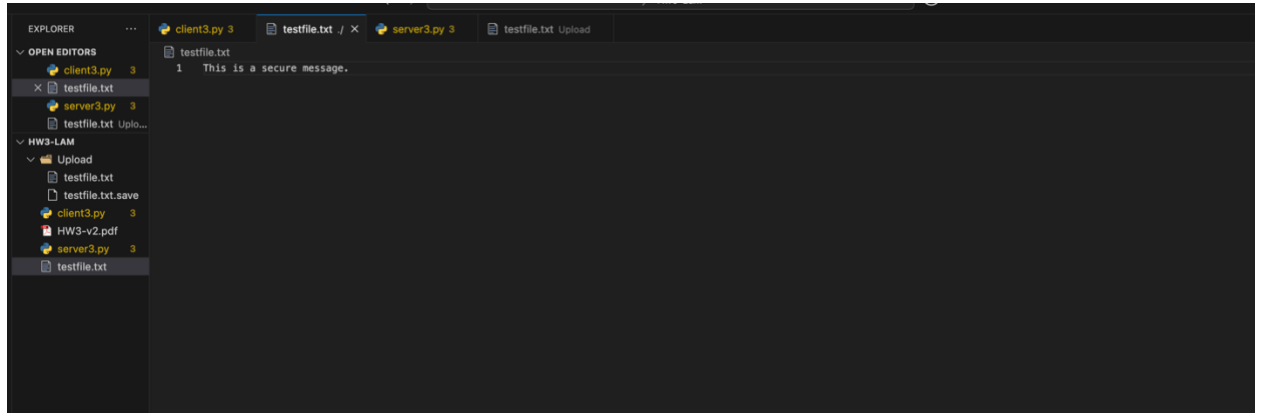
- a. Both client and server run on `localhost` using port `6000`.
- b. The server always sends its RSA public key first before any AES exchange.
- c. The client generates a new RSA key pair every run for simplicity.
- d. Files are small enough to be transmitted in one buffer; no chunking logic is needed.
- e. AES uses a 256-bit key with CFB mode and random IVs.
- f. Filename and content are UTF-8 encodable and can be reliably decoded.
- g. Signature verification uses SHA-256 hashing and PSS padding.

2. Authentication and Integrity Measures

- a. RSA Key Exchange: Server sends public key; client encrypts AES key with it.
- b. AES Encryption: File content and metadata are encrypted with AES.
- c. Digital Signature: Client signs SHA-256 hash of file with private key.
- d. Signature Verification: Server verifies using client's public key.
- e. Tamper Detection: Modified file fails hash comparison and verification.

3. Test Results

a. Valid file Test



Run python3 server.py

```
jasonlam@tris-air Hw3-Lam % python3 server3.py
Server: Listening on 127.0.0.1:6000...
```

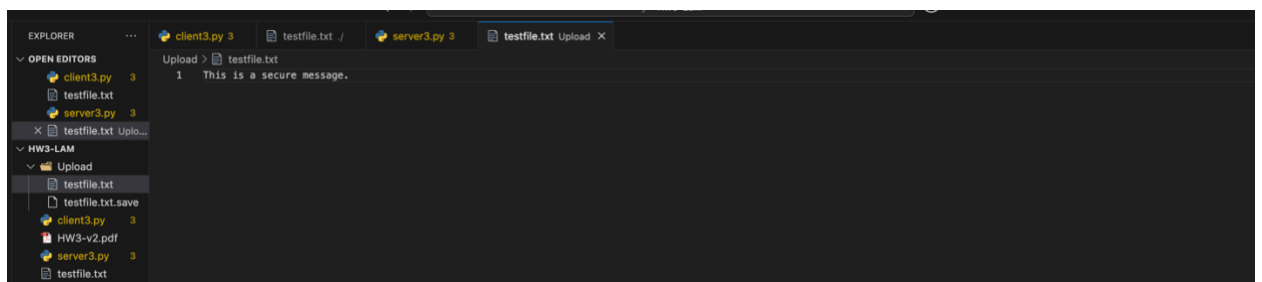
Run python3 client3.py

Then type testfile.txt

```
Enter the file name (add path if necessary): testfile.txt
Client: Connected to server (127.0.0.1, 6000)
>> Client: File 'testfile.txt' uploaded successfully.
jasonlam@tris-air Hw3-Lam %
```

Result

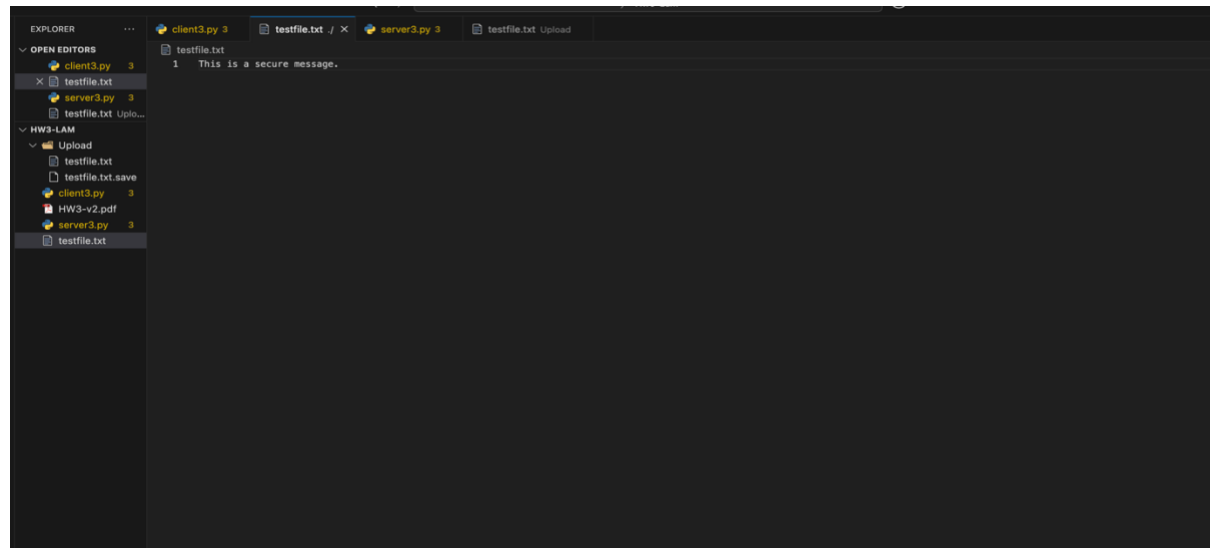
```
jasonlam@tris-air Hw3-Lam % python3 server3.py
Server: Listening on 127.0.0.1:6000...
Server: Connection accepted from ('127.0.0.1', 56192)
Server: File saved to 'Upload/testfile.txt'.
>> You may now edit the file before verification. Press Enter when ready...
✓ Server: Signature verified. File is authentic and unmodified.
```



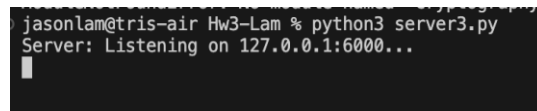
This show the file is verified and is authentic in the upload folder

b. Tampered File Test

This is the normal testfile.txt file

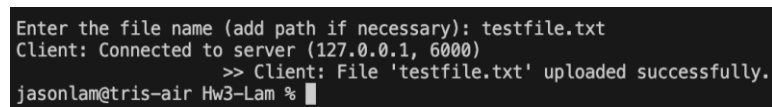


Run python3 server.py

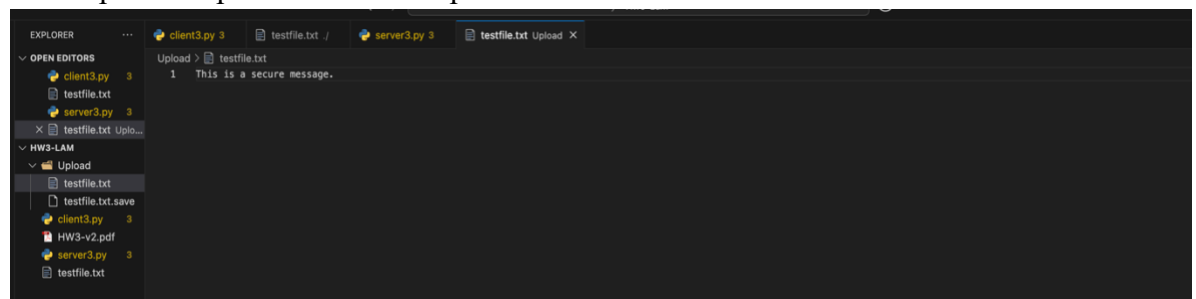


Run python3 client3.py

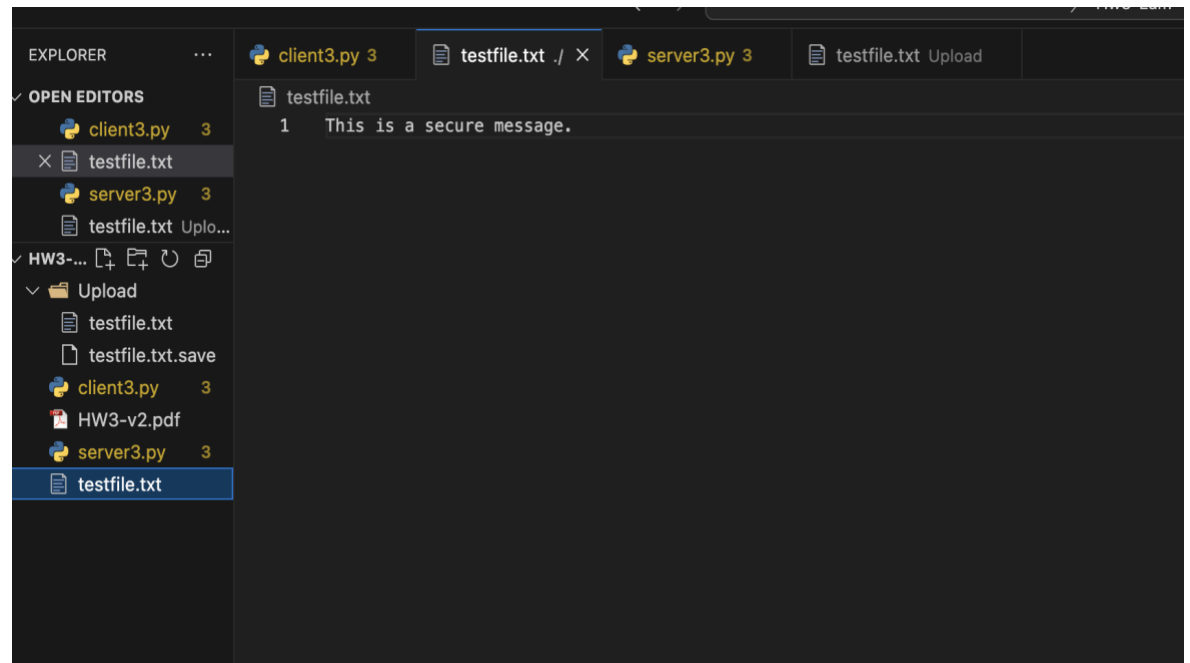
Then type testfile.txt



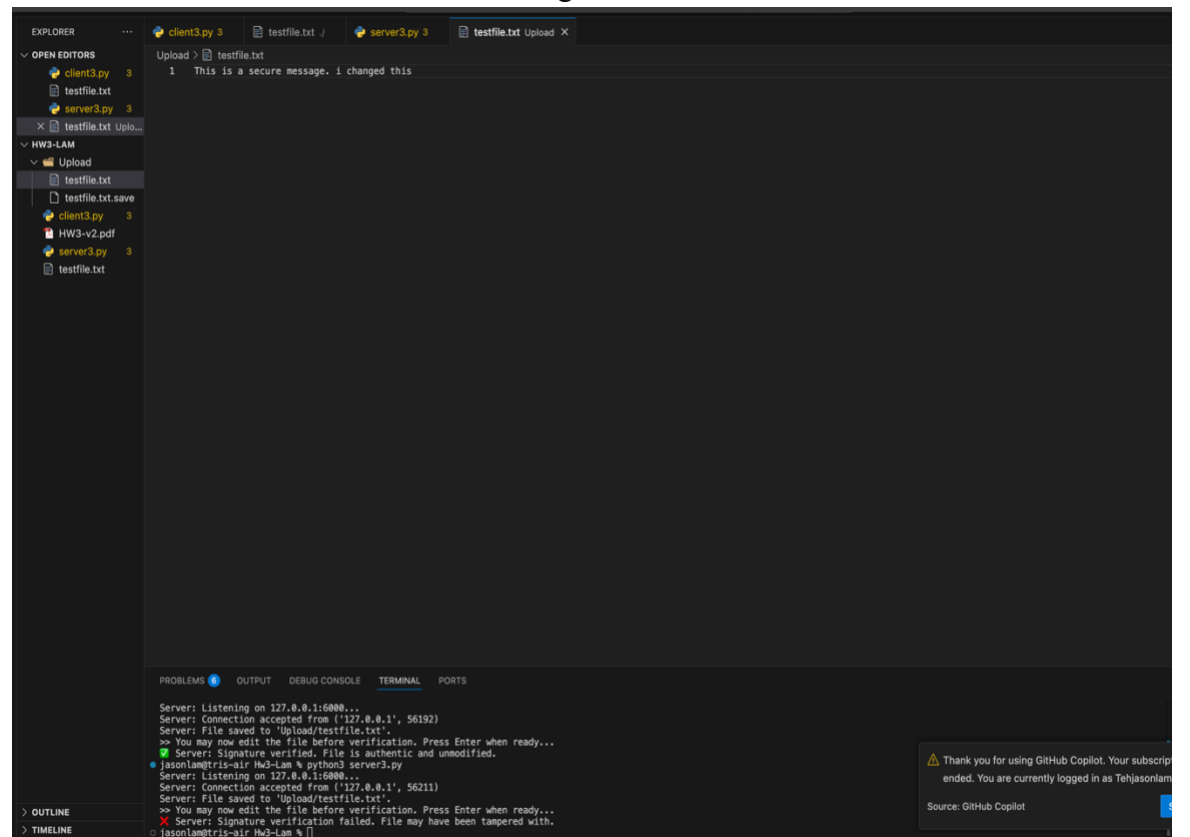
Then open the upload folder and open testfile.txt



Edit the testfile.txt



Press Enter in the Terminal for Server and get result



As you can see in the file I put “i changed this”. Then, at the bottom terminal The file was tempered.

4. Files Submitted

- a. client3.py
- b. server3.py
- c. testfile.txt (orginal)
- d. HW3.pdf
- e. Upload Folder with the testfile.txt file